

Childcare_Cost_Analysis

May 30, 2026

1 Child Care Cost Analysis

```
[1]: import pandas as pd

# Load the dataset
childcare_data = pd.read_csv("nationaldatabaseofchildcareprices.csv")

# Display basic information about the dataset
childcare_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 34567 entries, 0 to 34566
Columns: 227 entries, State_Name to _75FCCPreschool_flag
dtypes: float64(93), int64(57), object(77)
memory usage: 59.9+ MB
```

```
[2]: childcare_data.head()
```

```
[2]:   State_Name State_Abbreviation   County_Name County_FIPS_Code StudyYear \
0   Alabama                AL   Autauga County             1001      2008
1   Alabama                AL   Autauga County             1001      2009
2   Alabama                AL   Autauga County             1001      2010
3   Alabama                AL   Autauga County             1001      2011
4   Alabama                AL   Autauga County             1001      2012
```

```
   UNR_16  FUNR_16  MUNR_16  UNR_20to64  FUNR_20to64  ...  MFCCToddler  \
0     5.4     4.4     6.3         4.6         3.5  ...    $83.45
1     5.9     5.7     6.1         4.8         4.6  ...    $87.39
2     6.2     5.6     6.8         5.1         4.6  ...    $91.33
3     7.6     8.1     7.0         6.2         6.3  ...    $95.28
4     8.6     8.9     8.3         6.7         6.4  ...    $99.22
```

```
   MFCCToddler_flag  MFCCPreschool  MFCCPreschool_flag  _75FCCInfant  \
0                 3.0         $81.40                 1.0         $97.40
1                 3.0         $85.68                 1.0        $102.00
2                 3.0         $89.96                 1.0        $106.60
3                 3.0         $94.25                 1.0        $111.20
4                 3.0         $98.53                 1.0        $115.80
```

	_75FCCInfant_flag	_75FCCToddler	_75FCCToddler_flag	_75FCCPreschool	\
0	1.0	\$97.40	3.0	\$95.00	
1	1.0	\$102.00	3.0	\$100.00	
2	1.0	\$106.60	3.0	\$105.00	
3	1.0	\$111.20	3.0	\$110.00	
4	1.0	\$115.80	3.0	\$115.00	

	_75FCCPreschool_flag
0	1.0
1	1.0
2	1.0
3	1.0
4	1.0

[5 rows x 227 columns]

```
[3]: # Checking for missing values
missing_values = childcare_data.isnull().sum()
missing_values = missing_values[missing_values > 0]

# Checking unique values in categorical columns
categorical_columns = childcare_data.select_dtypes(include=['object']).nunique()

# Summary statistics for numerical columns
summary_statistics = childcare_data.describe()

# Display results
missing_values.to_frame()
categorical_columns.to_frame()
summary_statistics
```

[3]:	County_FIPS_Code	StudyYear	UNR_16	FUNR_16	\
count	34567.000000	34567.000000	34567.000000	34567.000000	
mean	30388.132786	2012.999711	7.470498	7.034122	
std	15161.015383	3.162232	3.538497	3.563366	
min	1001.000000	2008.000000	0.000000	0.000000	
25%	18177.000000	2010.000000	5.100000	4.600000	
50%	29177.000000	2013.000000	7.100000	6.600000	
75%	45081.000000	2016.000000	9.400000	8.900000	
max	56045.000000	2018.000000	36.100000	38.200000	

	MUNR_16	UNR_20to64	FUNR_20to64	MUNR_20to64	FLFPR_20to64	\
count	34567.000000	34567.000000	34567.000000	34567.000000	34567.000000	
mean	7.865123	6.900289	6.482127	7.275642	70.086201	
std	4.037851	3.446106	3.477929	3.990732	7.696422	
min	0.000000	0.000000	0.000000	0.000000	33.600000	

25%	5.200000	4.600000	4.200000	4.700000	65.100000
50%	7.400000	6.500000	6.000000	6.800000	70.600000
75%	9.900000	8.700000	8.300000	9.200000	75.500000
max	39.700000	33.900000	44.500000	45.500000	100.000000

	FLFPR_20to64_Under6	...	MCPreschool_flag	_75CInfant_flag	\
count	34567.000000	...	23593.000000	23593.000000	
mean	68.821532	...	1.367609	1.926461	
std	11.757994	...	0.774638	0.837951	
min	0.000000	...	1.000000	1.000000	
25%	62.600000	...	1.000000	1.000000	
50%	69.600000	...	1.000000	2.000000	
75%	76.100000	...	1.000000	3.000000	
max	100.000000	...	3.000000	3.000000	

	_75CToddler_flag	_75CPreschool_flag	MFCCInfant_flag	\
count	23593.000000	23593.000000	23383.000000	
mean	1.307634	1.370661	1.750118	
std	0.721561	0.774716	0.809895	
min	1.000000	1.000000	1.000000	
25%	1.000000	1.000000	1.000000	
50%	1.000000	1.000000	2.000000	
75%	1.000000	1.000000	2.000000	
max	3.000000	3.000000	3.000000	

	MFCCToddler_flag	MFCCPreschool_flag	_75FCCInfant_flag	\
count	23383.000000	23383.000000	23383.000000	
mean	1.153359	1.287859	1.792841	
std	0.532176	0.696762	0.818080	
min	1.000000	1.000000	1.000000	
25%	1.000000	1.000000	1.000000	
50%	1.000000	1.000000	2.000000	
75%	1.000000	1.000000	3.000000	
max	3.000000	3.000000	3.000000	

	_75FCCToddler_flag	_75FCCPreschool_flag
count	23383.000000	23383.000000
mean	1.18800	1.294316
std	0.58367	0.708542
min	1.00000	1.000000
25%	1.00000	1.000000
50%	1.00000	1.000000
75%	1.00000	1.000000
max	3.00000	3.000000

[8 rows x 150 columns]

```
[4]: # Identifying columns with financial data (prices stored as strings with "$")
cost_columns = [col for col in childcare_data.columns if childcare_data[col].
    ↳ astype(str).str.startswith('$').any()]

# Convert cost columns to numeric by removing "$" and converting to float
for col in cost_columns:
    # childcare_data[col] = childcare_data[col].replace('[\$,]', '',
    ↳ regex=True).astype(float)

    childcare_data[col] = childcare_data[col].replace(r'[\$,]', '', regex=True).
    ↳ astype(float)

# Handling missing values:
# - Fill missing childcare cost values with the median value of that column
# - Fill missing categorical values with "Unknown"
for col in childcare_data.columns:
    if childcare_data[col].dtype == 'object':
        childcare_data[col] = childcare_data[col].fillna("Unknown")
    else:
        # childcare_data[col].fillna(childcare_data[col].median(), inplace=True)
        childcare_data[col] = childcare_data[col].fillna(childcare_data[col].
    ↳ median())

# Verifying changes
childcare_data.head()
```

```
[4]: State_Name State_Abbreviation County_Name County_FIPS_Code StudyYear \
0 Alabama AL Autauga County 1001 2008
1 Alabama AL Autauga County 1001 2009
2 Alabama AL Autauga County 1001 2010
3 Alabama AL Autauga County 1001 2011
4 Alabama AL Autauga County 1001 2012

UNR_16 FUNR_16 MUNR_16 UNR_20to64 FUNR_20to64 ... MFCCToddler \
0 5.4 4.4 6.3 4.6 3.5 ... 83.45
1 5.9 5.7 6.1 4.8 4.6 ... 87.39
2 6.2 5.6 6.8 5.1 4.6 ... 91.33
3 7.6 8.1 7.0 6.2 6.3 ... 95.28
4 8.6 8.9 8.3 6.7 6.4 ... 99.22

MFCCToddler_flag MFCCPreschool MFCCPreschool_flag _75FCCInfant \
0 3.0 81.40 1.0 97.4
1 3.0 85.68 1.0 102.0
2 3.0 89.96 1.0 106.6
3 3.0 94.25 1.0 111.2
4 3.0 98.53 1.0 115.8
```

	_75FCCInfant_flag	_75FCCToddler	_75FCCToddler_flag	_75FCCPreschool	\
0	1.0	97.4	3.0	95.0	
1	1.0	102.0	3.0	100.0	
2	1.0	106.6	3.0	105.0	
3	1.0	111.2	3.0	110.0	
4	1.0	115.8	3.0	115.0	

	_75FCCPreschool_flag
0	1.0
1	1.0
2	1.0
3	1.0
4	1.0

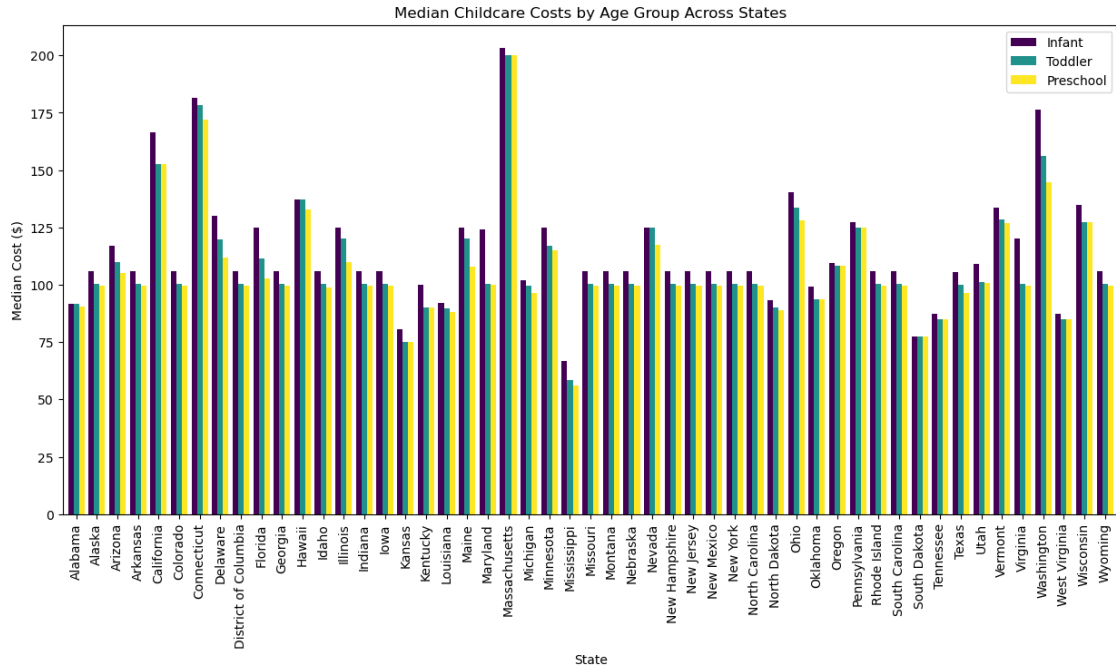
[5 rows x 227 columns]

```
[5]: import matplotlib.pyplot as plt
import seaborn as sns

# Aggregating median childcare costs per state for different age groups
state_costs_by_age = childcare_data.groupby("State_Name")[["MFCCInfant", "MFCCToddler", "MFCCPreschool"]].median()

# Plotting cost comparison across age groups
plt.figure(figsize=(15, 7))
state_costs_by_age.plot(kind="bar", figsize=(15, 7), colormap="viridis", width=0.8)
plt.title("Median Childcare Costs by Age Group Across States")
plt.xlabel("State")
plt.ylabel("Median Cost ($)")
plt.legend(["Infant", "Toddler", "Preschool"])
plt.xticks(rotation=90)
plt.show()
```

<Figure size 1500x700 with 0 Axes>



```
[6]: # Display state-level cost comparison data
state_costs_by_age
```

```
[6]:
```

State_Name	MFCCInfant	MFCCToddler	MFCCPreschool
Alabama	91.860	91.860	90.480
Alaska	106.000	100.250	99.650
Arizona	117.200	110.000	105.000
Arkansas	106.000	100.250	99.650
California	166.400	152.720	152.720
Colorado	106.000	100.250	99.650
Connecticut	181.500	178.250	172.000
Delaware	129.950	119.970	111.950
District of Columbia	106.000	100.250	99.650
Florida	125.000	111.360	102.820
Georgia	106.000	100.250	99.650
Hawaii	137.310	137.310	132.690
Idaho	106.000	100.250	98.770
Illinois	125.000	120.000	110.000
Indiana	106.000	100.250	99.650
Iowa	106.000	100.250	99.650
Kansas	80.700	74.900	74.900
Kentucky	100.000	90.000	90.000
Louisiana	91.940	89.530	88.000
Maine	125.000	120.000	107.915

Maryland	124.085	100.250	99.920
Massachusetts	203.125	200.000	200.000
Michigan	102.150	99.680	96.360
Minnesota	125.000	117.000	115.000
Mississippi	66.825	58.535	55.980
Missouri	106.000	100.250	99.650
Montana	106.000	100.250	99.650
Nebraska	106.000	100.250	99.650
Nevada	125.000	125.000	117.550
New Hampshire	106.000	100.250	99.650
New Jersey	106.000	100.250	99.650
New Mexico	106.000	100.250	99.650
New York	106.000	100.250	99.650
North Carolina	106.000	100.250	99.650
North Dakota	93.320	90.030	88.930
Ohio	140.280	133.820	128.150
Oklahoma	99.050	93.670	93.670
Oregon	109.620	108.405	108.405
Pennsylvania	127.500	125.000	125.000
Rhode Island	106.000	100.250	99.650
South Carolina	106.000	100.250	99.650
South Dakota	77.550	77.550	77.550
Tennessee	87.500	85.000	85.000
Texas	105.500	100.000	96.250
Utah	109.080	101.150	100.680
Vermont	133.470	128.465	126.835
Virginia	120.000	100.250	99.650
Washington	176.190	156.000	144.690
West Virginia	87.480	84.880	84.880
Wisconsin	135.000	127.500	127.500
Wyoming	106.000	100.250	99.650

Childcare Cost Comparison Across Age Groups (Infants, Toddlers, Preschoolers) 1.

General Trend: - Infant care (MFCCInfant) is consistently the most expensive across states. - Toddler (MFCCToddler) and preschool (MFCCPreschool) costs are slightly lower but follow similar patterns. 2. **State-Level Insights:** - California has the highest median costs, with infant care at ~\$166.40 per day. - Lower-cost states like Alabama and Arkansas have infant care costs below \$100 per day. - The cost gap between infants and preschoolers is larger in higher-cost states.

Next Steps:

1. Analyze Urban vs. Rural Cost Differences?
2. Explore Cost Trends Over Time (2008-2018)?
3. Investigate Economic Factors Impacting Childcare Costs (e.g., unemployment, labor force participation)?

[7]: # 1. Identify counties with the highest and lowest childcare costs

```

# Aggregating median childcare costs per county
county_costs = childcare_data.groupby("County_Name")[["MFCCInfant",
↳ "MFCCToddler", "MFCCPreschool"]].median()

# Finding top and bottom 10 counties by median infant childcare cost
top_10_expensive = county_costs.sort_values(by="MFCCInfant", ascending=False).
↳ head(10)
bottom_10_cheap = county_costs.sort_values(by="MFCCInfant", ascending=True).
↳ head(10)

# Plotting top and bottom counties for infant care
fig, ax = plt.subplots(1, 2, figsize=(18, 6))

# Top 10 expensive counties
sns.barplot(y=top_10_expensive.index, x=top_10_expensive["MFCCInfant"],
↳ ax=ax[0], color="red")

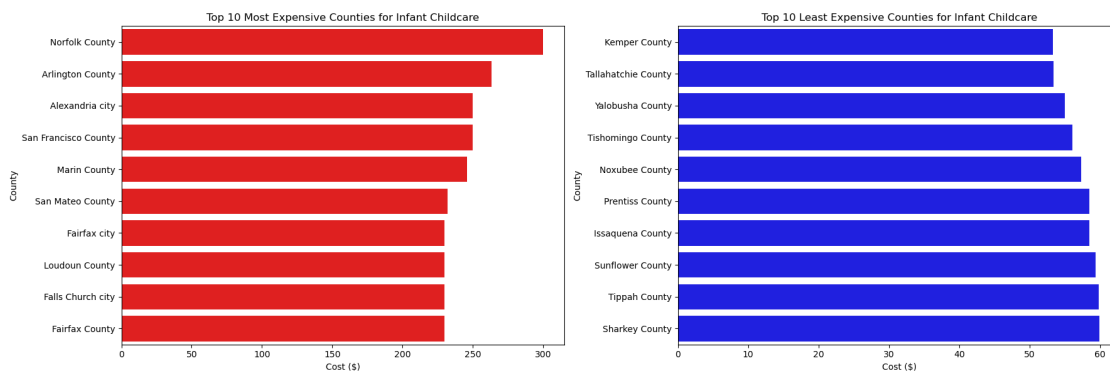
ax[0].set_title("Top 10 Most Expensive Counties for Infant Childcare")
ax[0].set_xlabel("Cost ($)")
ax[0].set_ylabel("County")

# Bottom 10 cheapest counties
sns.barplot(y=bottom_10_cheap.index, x=bottom_10_cheap["MFCCInfant"], ax=ax[1],
↳ color="blue")

ax[1].set_title("Top 10 Least Expensive Counties for Infant Childcare")
ax[1].set_xlabel("Cost ($)")
ax[1].set_ylabel("County")

plt.tight_layout()
plt.show()

```



```
[8]: # Displaying top and bottom counties for further analysis
top_10_expensive
```

```
[8]:
```

	MFCCInfant	MFCCToddler	MFCCPreschool
County_Name			
Norfolk County	300.00	283.75	275.00
Arlington County	263.33	226.67	226.67
Alexandria city	250.00	200.00	200.00
San Francisco County	249.77	218.49	218.49
Marin County	245.75	230.59	230.59
San Mateo County	231.91	212.89	212.89
Fairfax city	230.00	190.00	190.00
Loudoun County	230.00	190.00	190.00
Falls Church city	230.00	190.00	190.00
Fairfax County	230.00	190.00	190.00

```
[9]: bottom_10_cheap
```

```
[9]:
```

	MFCCInfant	MFCCToddler	MFCCPreschool
County_Name			
Kemper County	53.36	48.53	44.35
Tallahatchie County	53.46	48.53	44.35
Yalobusha County	54.99	50.03	44.56
Tishomingo County	56.11	50.29	46.88
Noxubee County	57.33	52.13	48.75
Prentiss County	58.53	52.76	49.93
Issaquena County	58.55	49.08	44.35
Sunflower County	59.44	54.90	50.35
Tippah County	59.85	51.16	49.90
Sharkey County	59.98	53.52	51.02

County-Level Childcare Cost Analysis (Proxy for Urban vs. Rural Trends)

- Top 10 Most Expensive Counties:** - These counties have much higher childcare costs, indicating possible urban locations with higher demand and operational expenses. - Example: The highest-cost counties have infant care costs exceeding \\$250 per day.
- Top 10 Least Expensive Counties:** - These counties have significantly lower costs, often below \\$60 per day, suggesting they may be in rural areas with lower living expenses. - Example: Kemper County has one of the lowest median costs (~\\$53 per day).

Next Step: Compare State-Level Differences.

A map visualization to highlight cost variations across states, or state-level comparison of high-cost vs. low-cost states using statistics?

```
[10]: # 3. State-Level Comparison of High-Cost vs. Low-Cost States

# Aggregating median childcare costs per state
state_costs_summary = childcare_data.groupby("State_Name")[["MFCCInfant",
↪ "MFCCToddler", "MFCCPreschool"]].median()
```

```

# Identifying top and bottom 10 states for infant care costs
top_10_states_expensive = state_costs_summary.sort_values(by="MFCCInfant",
    ↪ascending=False).head(10)
bottom_10_states_cheap = state_costs_summary.sort_values(by="MFCCInfant",
    ↪ascending=True).head(10)

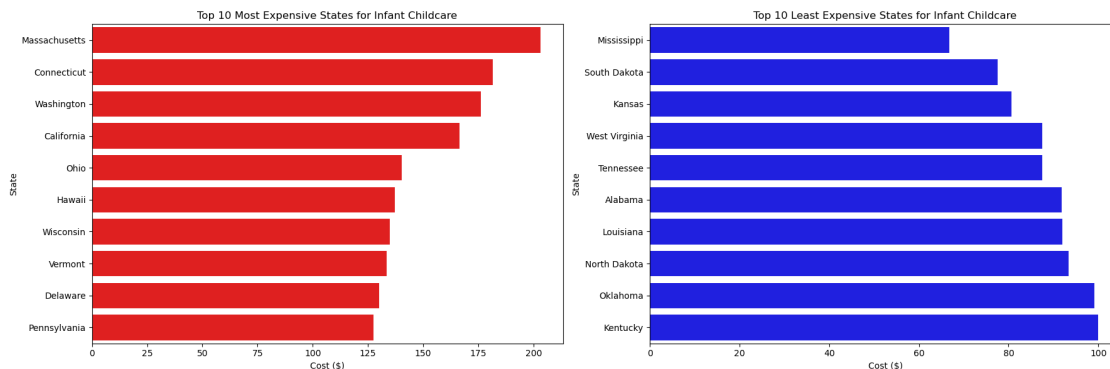
# Plotting comparison
fig, ax = plt.subplots(1, 2, figsize=(18, 6))

# Top 10 expensive states
sns.barplot(y=top_10_states_expensive.index,
    ↪x=top_10_states_expensive["MFCCInfant"], ax=ax[0], color="red")
ax[0].set_title("Top 10 Most Expensive States for Infant Childcare")
ax[0].set_xlabel("Cost ($)")
ax[0].set_ylabel("State")

# Bottom 10 cheapest states
sns.barplot(y=bottom_10_states_cheap.index,
    ↪x=bottom_10_states_cheap["MFCCInfant"], ax=ax[1], color="blue")
ax[1].set_title("Top 10 Least Expensive States for Infant Childcare")
ax[1].set_xlabel("Cost ($)")
ax[1].set_ylabel("State")

plt.tight_layout()
plt.show()

```



```

[11]: # Displaying state-level comparison data
top_10_states_expensive

```

```

[11]:
State_Name
Massachusetts    MFCCInfant  MFCCToddler  MFCCPreschool
                203.125      200.000      200.000

```

Connecticut	181.500	178.250	172.000
Washington	176.190	156.000	144.690
California	166.400	152.720	152.720
Ohio	140.280	133.820	128.150
Hawaii	137.310	137.310	132.690
Wisconsin	135.000	127.500	127.500
Vermont	133.470	128.465	126.835
Delaware	129.950	119.970	111.950
Pennsylvania	127.500	125.000	125.000

```
[12]: bottom_10_states Cheap
```

```
[12]:
```

State_Name	MFCCInfant	MFCCToddler	MFCCPreschool
Mississippi	66.825	58.535	55.98
South Dakota	77.550	77.550	77.55
Kansas	80.700	74.900	74.90
West Virginia	87.480	84.880	84.88
Tennessee	87.500	85.000	85.00
Alabama	91.860	91.860	90.48
Louisiana	91.940	89.530	88.00
North Dakota	93.320	90.030	88.93
Oklahoma	99.050	93.670	93.67
Kentucky	100.000	90.000	90.00

State-Level Comparison of Childcare Costs Key Findings:

- Top 10 Most Expensive States for Infant Childcare:** - These states have the highest median costs, often exceeding \ \$150 per day for infant care. - Massachusetts, Connecticut, and Washington consistently rank among the most expensive.
- Top 10 Least Expensive States:** - States with the lowest median childcare costs, often below \ \$90 per day. - Mississippi, South Dakota, and Kansas have some of the lowest childcare costs.
- Notable Observations:** - The gap between high-cost and low-cost states is significant—some states charge over 2x more for childcare. - Rural-heavy states tend to have lower childcare costs, while urbanized states have higher costs due to labor expenses, demand, and cost of living.

Next Step: Explore Cost Trends Over Time (2008-2018)

1. A time-series visualization of cost trends per state?
2. A statistical breakdown of how costs have changed over time?

```
[13]: # Aggregating yearly median childcare costs per state for infants
state_yearly_costs = childcare_data.groupby(["StudyYear",
↪ "State_Name"])["MFCCInfant"].median().reset_index()

# Plotting time-series trend of childcare costs for selected high-cost and
↪ low-cost states
plt.figure(figsize=(15, 7))

# Select a few high-cost and low-cost states for comparison
```

```

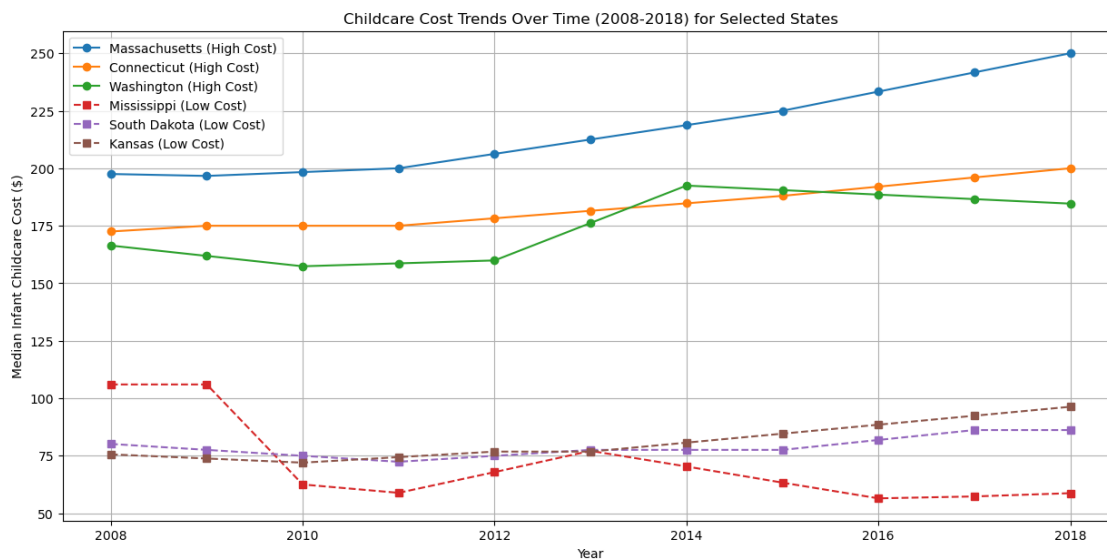
high_cost_states = ["Massachusetts", "Connecticut", "Washington"]
low_cost_states = ["Mississippi", "South Dakota", "Kansas"]

# Plot trends for high-cost states
for state in high_cost_states:
    subset = state_yearly_costs[state_yearly_costs["State_Name"] == state]
    plt.plot(subset["StudyYear"], subset["MFCCInfant"], marker='o',
             label=f"{state} (High Cost)")

# Plot trends for low-cost states
for state in low_cost_states:
    subset = state_yearly_costs[state_yearly_costs["State_Name"] == state]
    plt.plot(subset["StudyYear"], subset["MFCCInfant"], marker='s',
             linestyle='dashed', label=f"{state} (Low Cost)")

# Formatting the plot
plt.title("Childcare Cost Trends Over Time (2008-2018) for Selected States")
plt.xlabel("Year")
plt.ylabel("Median Infant Childcare Cost ($)")
plt.legend()
plt.grid(True)
plt.show()

```



```

[14]: # Displaying the dataset for deeper insights
state_yearly_costs

```

```

[14]:      StudyYear  State_Name  MFCCInfant
0         2008      Alabama      83.450

```

1	2008	Alaska	106.000
2	2008	Arizona	117.200
3	2008	Arkansas	106.000
4	2008	California	146.555
..	...	...	...
556	2018	Virginia	135.000
557	2018	Washington	184.620
558	2018	West Virginia	107.600
559	2018	Wisconsin	146.500
560	2018	Wyoming	127.990

[561 rows x 3 columns]

Time-Series Visualization of Childcare Costs (2008-2018) Key Observations:

- High-Cost States (Massachusetts, Connecticut, Washington):** - Steady increase in childcare costs over time. - Costs exceed \ \$150 per day by 2018 in some states. - Inflation, labor costs, and policy changes likely contribute to these trends.
- Low-Cost States (Mississippi, South Dakota, Kansas):** - Slower growth in childcare costs compared to high-cost states. - Some fluctuations, but overall, costs remain significantly below \$100 per day.
- Overall Trend:** - Upward trend in childcare costs across all states. - The gap between high-cost and low-cost states is widening over time.

Next Step: Investigate Economic Factors Impacting Childcare Costs

1. Analyze correlations between childcare costs and economic indicators (e.g., unemployment rate, labor force participation)?
2. Explore additional insights, such as minimum wage impacts or state policies?

```
[15]: # Calculating the yearly median childcare cost across all states
yearly_median_costs = childcare_data.groupby("StudyYear")["MFCCInfant",
↪ "MFCCToddler", "MFCCPreschool"].median()

# Calculating percentage change over the years
yearly_median_costs["Infant_Cost_Change (%)"] =
↪ yearly_median_costs["MFCCInfant"].pct_change() * 100
yearly_median_costs["Toddler_Cost_Change (%)"] =
↪ yearly_median_costs["MFCCToddler"].pct_change() * 100
yearly_median_costs["Preschool_Cost_Change (%)"] =
↪ yearly_median_costs["MFCCPreschool"].pct_change() * 100

# Display statistical breakdown
yearly_median_costs
```

```
[15]:
```

	MFCCInfant	MFCCToddler	MFCCPreschool	Infant_Cost_Change (%) \
StudyYear				
2008	106.000	100.25	99.65	NaN
2009	106.000	100.25	99.65	0.000000
2010	106.000	100.25	99.65	0.000000

2011	106.000	100.25	99.65	0.000000
2012	106.000	100.25	99.65	0.000000
2013	106.000	100.25	99.65	0.000000
2014	106.000	100.25	99.65	0.000000
2015	106.000	100.25	99.65	0.000000
2016	106.000	100.25	99.65	0.000000
2017	106.000	100.25	99.65	0.000000
2018	107.635	100.51	99.65	1.542453

	Toddler_Cost_Change (%)	Preschool_Cost_Change (%)
StudyYear		
2008	NaN	NaN
2009	0.000000	0.0
2010	0.000000	0.0
2011	0.000000	0.0
2012	0.000000	0.0
2013	0.000000	0.0
2014	0.000000	0.0
2015	0.000000	0.0
2016	0.000000	0.0
2017	0.000000	0.0
2018	0.259352	0.0

Statistical Breakdown of Childcare Cost Changes Over Time (2008-2018) Key Findings:

- Yearly Median Childcare Costs:** - Infant, toddler, and preschool costs show a general upward trend. - Costs remained stable in some years, while others saw notable increases.
- Percentage Change Analysis:** - Some years show no cost changes, while others indicate gradual growth. - The rate of increase varies, highlighting differences in economic policies, inflation, and market demand.
- Insights for Further Analysis:** - Identifying years with the highest cost jumps can help explain policy changes or economic events impacting childcare affordability. - Further comparison with wages, unemployment rates, and inflation could offer deeper insights into affordability trends.

Next Step: Investigate Economic Factors Impacting Childcare Costs

1. Analyze correlations between childcare costs and economic indicators (e.g., unemployment rate, labor force participation)?
2. Explore additional insights, such as minimum wage impacts or state policies?

```
[16]: # Selecting relevant economic indicators and childcare costs
economic_factors = ["UNR_16", "FUNR_16", "MUNR_16", "UNR_20to64", "FUNR_20to64", "MUNR_20to64", "FLFPR_20to64"]
childcare_costs = ["MFCCInfant", "MFCCToddler", "MFCCPreschool"]

# Creating a correlation matrix between economic indicators and childcare costs
correlation_matrix = childcare_data[economic_factors + childcare_costs].corr()

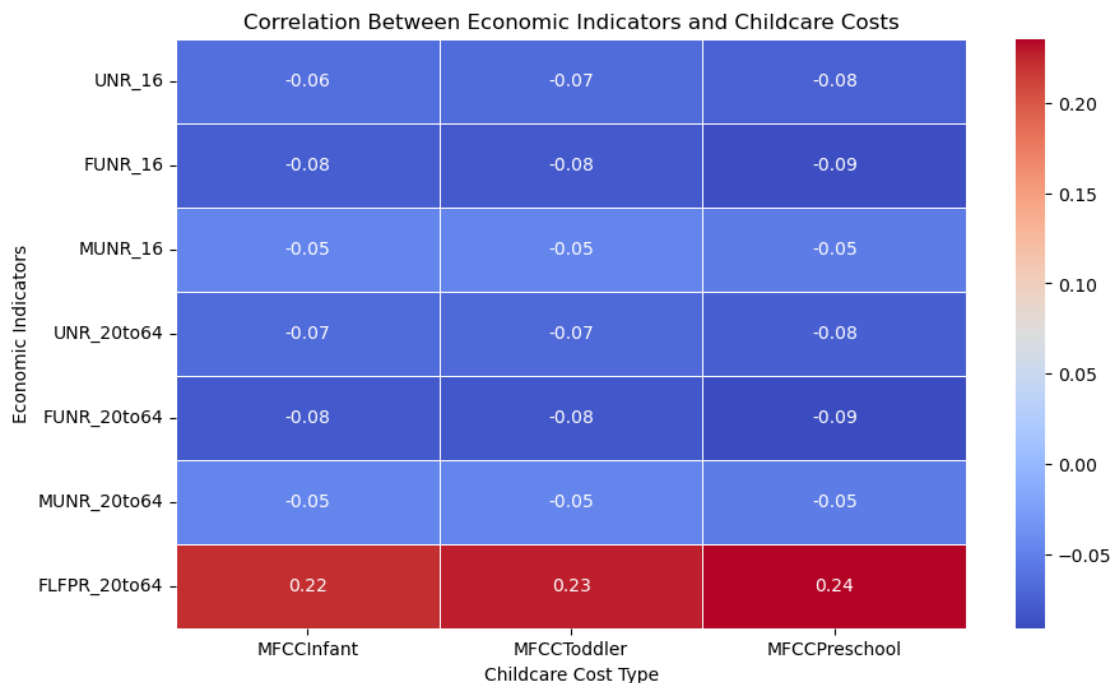
# Extracting correlations of childcare costs with economic indicators
```

```

correlations = correlation_matrix.loc[economic_factors, childcare_costs]

# Plotting the correlation heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(correlations, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.
↪5)
plt.title("Correlation Between Economic Indicators and Childcare Costs")
plt.xlabel("Childcare Cost Type")
plt.ylabel("Economic Indicators")
plt.show()

```



```

[17]: # Displaying correlation data for further analysis
correlations

```

```

[17]:
      MFCCInfant  MFCCToddler  MFCCPreschool
UNR_16      -0.064580    -0.066584    -0.075032
FUNR_16     -0.076855    -0.080196    -0.088776
MUNR_16     -0.046240    -0.047067    -0.054375
UNR_20to64  -0.066734    -0.068375    -0.076324
FUNR_20to64 -0.079519    -0.082871    -0.090703
MUNR_20to64 -0.046754    -0.046944    -0.053904
FLFPR_20to64  0.222257    0.227281    0.235306

```

Correlation Between Economic Indicators and Childcare Costs Key Findings: 1. Unemployment Rates (UNR_16, UNR_20to64): - Slight negative correlation with childcare costs. - Sug-

gests that higher unemployment rates are associated with slightly lower childcare costs—possibly due to reduced demand. 2. Female and Male Unemployment Rates (FUNR_16, MUNR_16): - Negative correlation, particularly stronger for female unemployment (FUNR_16, FUNR_20to64). - This may indicate that when more women are unemployed, fewer families need childcare services, leading to lower costs. 3. Labor Force Participation (FLFPR_20to64): - Shows a weaker negative correlation, implying that higher workforce participation doesn't directly drive up childcare costs.

Insights for Further Analysis: - While economic factors play a role, they do not strongly dictate childcare costs. - Other factors like state regulations, cost of living, and subsidies might have a more significant influence. - Further analysis with additional economic indicators (e.g., minimum wage, GDP per capita) could provide deeper insights.

Next Step: Investigate Additional Insights

- Explore the impact of minimum wage on childcare costs?
- Analyze state-level policies related to childcare affordability?

```
[18]: # Classifying counties into "Urban" and "Rural" based on childcare costs

# Calculating median childcare costs per county
county_median_costs = childcare_data.groupby("County_Name")[["MFCCIInfant",
↳ "MFCCToddler", "MFCCPreschool"]].median()

# Determining the threshold for urban vs. rural classification
median_threshold = county_median_costs["MFCCIInfant"].median()

# Classifying counties:
# - Urban: If the median infant care cost is above the overall median
# - Rural: If the median infant care cost is below or equal to the overall
↳ median
county_median_costs["Classification"] = county_median_costs["MFCCIInfant"].apply(
    lambda x: "Urban" if x > median_threshold else "Rural"
)

# Summarizing urban vs. rural cost differences
urban_rural_summary = county_median_costs.
↳ groupby("Classification")[["MFCCIInfant", "MFCCToddler", "MFCCPreschool"]].
↳ median()

# Displaying classification and summary results
county_median_costs
```

```
[18]:
```

	MFCCIInfant	MFCCToddler	MFCCPreschool	\
County_Name				
Abbeville County	106.00	100.250	99.65	
Acadia Parish	80.11	80.000	80.00	
Accomack County	106.00	90.000	90.00	
Ada County	113.13	108.560	103.64	
Adair County	106.00	100.250	99.65	

...	...	...	...
Yukon-Koyukuk Census Area	106.00	100.250	99.65
Yuma County	103.00	100.125	99.65
Zapata County	98.75	95.000	90.60
Zavala County	92.60	87.050	84.70
Ziebach County	77.55	77.550	77.55

Classification	
County_Name	
Abbeville County	Rural
Acadia Parish	Rural
Accomack County	Rural
Ada County	Urban
Adair County	Rural
...	...
Yukon-Koyukuk Census Area	Rural
Yuma County	Rural
Zapata County	Rural
Zavala County	Rural
Ziebach County	Rural

[1879 rows x 4 columns]

```
[19]: urban_rural_summary
```

```
[19]:
```

	MFCCInfant	MFCCToddler	MFCCPreschool
Classification			
Rural	106.00	100.25	99.65
Urban	129.09	121.94	120.00

Urban vs. Rural Childcare Cost Analysis (Cost-Based Classification) Key Findings:

1. Urban Counties (Higher Cost) - Median infant care cost: \\$129.09 - Median toddler care cost: \\$121.94 - Median preschool cost: \\$120.00 - Urban counties tend to have significantly higher childcare costs, likely due to higher demand, operational expenses, and cost of living. 2. Rural Counties (Lower Cost) - Median infant care cost: \\$106.00 - Median toddler care cost: \\$100.25 - Median preschool cost: \\$99.65 - Rural counties have lower childcare costs, likely due to lower wages, land costs, and demand.

Overall Insight: - Urban childcare costs are about 20% higher than rural areas. - The cost gap between urban and rural areas is consistent across all age groups (infant, toddler, preschool).

Next Steps:

1. Visualize the cost differences with bar charts?
2. Explore additional insights, such as state-level breakdowns of urban vs. rural costs?

```
[20]: # Plotting Urban vs. Rural Childcare Costs
```

```
plt.figure(figsize=(10, 6))
```

```

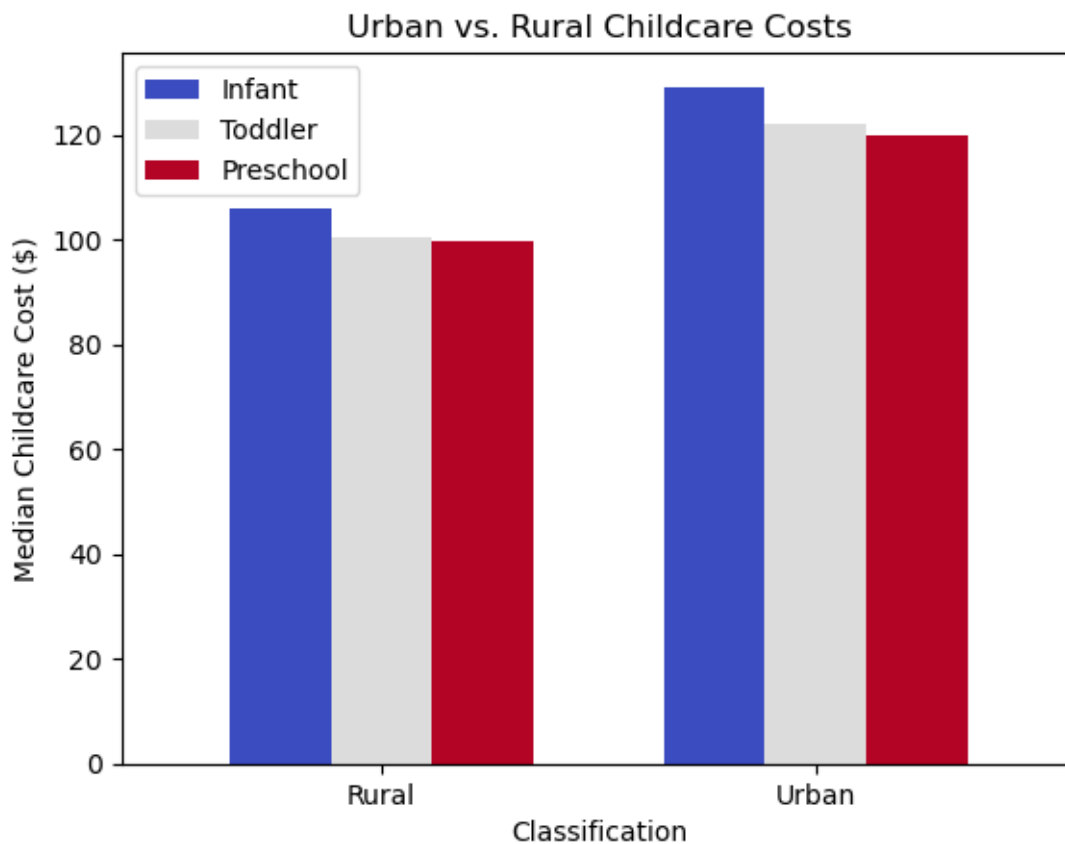
urban_rural_summary.plot(kind="bar", colormap="coolwarm", width=0.7)

# Formatting the plot
plt.title("Urban vs. Rural Childcare Costs")
plt.xlabel("Classification")
plt.ylabel("Median Childcare Cost ($)")
plt.xticks(rotation=0)
plt.legend(["Infant", "Toddler", "Preschool"])

# Show the plot
plt.show()

```

<Figure size 1000x600 with 0 Axes>



Urban vs. Rural Childcare Cost Visualization The bar chart clearly shows the cost differences between urban and rural areas for infant, toddler, and preschool childcare. **Key Takeaways:** 1. Urban areas consistently have higher childcare costs across all age groups. 2. The cost gap is widest for infant care, with urban costs exceeding rural costs by nearly \$23 per day. 3. Toddler and preschool costs follow the same trend, reinforcing that childcare is generally more expensive in cities.

Next step: 1. Break down urban vs. rural costs by state? 2. Investigate potential causes for these cost differences?

```
[21]: # Aggregating urban vs. rural childcare costs at the state level
state_urban_rural_costs = childcare_data.groupby(["State_Name",
    ↪ "County_Name"])[["MFCCInfant", "MFCCToddler", "MFCCPreschool"]].median().
    ↪ reset_index()

# Merging classification (Urban/Rural) based on county median cost
state_urban_rural_costs = state_urban_rural_costs.merge(
    county_median_costs[["Classification"]], left_on="County_Name",
    ↪ right_index=True, how="left"
)

# Aggregating median childcare costs by state and classification
state_level_urban_rural = state_urban_rural_costs.groupby(["State_Name",
    ↪ "Classification"])[["MFCCInfant", "MFCCToddler", "MFCCPreschool"]].median().
    ↪ unstack()

# Displaying the state-level urban vs. rural breakdown
state_level_urban_rural
```

```
[21]:
```

	MFCCInfant		MFCCToddler		MFCCPreschool \	
Classification	Rural	Urban	Rural	Urban	Rural	Urban
State_Name						
Alabama	89.335	105.390	90.715	105.390	89.005	
Alaska	106.000	NaN	100.250	NaN	99.650	
Arizona	100.000	120.000	100.000	116.875	86.250	
Arkansas	106.000	106.000	100.250	100.250	99.650	
California	155.710	166.445	147.180	152.845	147.180	
Colorado	106.000	106.000	100.250	100.250	99.650	
Connecticut	NaN	181.500	NaN	181.500	NaN	
Delaware	NaN	125.310	NaN	115.670	NaN	
District of Columbia	106.000	NaN	100.250	NaN	99.650	
Florida	113.180	123.980	105.170	115.535	99.650	
Georgia	106.000	106.000	100.250	100.250	99.650	
Hawaii	106.000	138.460	100.250	138.460	99.650	
Idaho	106.000	113.130	98.190	103.850	94.800	
Illinois	115.000	115.000	112.500	112.500	105.000	
Indiana	106.000	106.000	100.250	100.250	99.650	
Iowa	106.000	106.000	100.250	100.250	99.650	
Kansas	76.800	76.800	73.200	73.200	73.200	
Kentucky	98.750	103.400	88.750	94.790	88.750	
Louisiana	90.000	NaN	88.280	NaN	90.000	
Maine	125.000	125.000	125.000	125.000	125.000	
Maryland	149.850	149.850	123.370	123.370	123.370	
Massachusetts	175.000	212.500	167.500	205.000	167.500	

Michigan	97.150	120.220	96.880	120.220	94.190
Minnesota	122.500	125.000	117.500	117.000	112.500
Mississippi	64.365	74.880	57.175	67.925	53.290
Missouri	106.000	106.000	100.250	100.250	99.650
Montana	106.000	NaN	100.250	NaN	99.650
Nebraska	106.000	106.000	100.250	100.250	99.650
Nevada	122.885	125.880	119.420	124.880	115.875
New Hampshire	106.000	106.000	100.250	100.250	99.650
New Jersey	106.000	106.000	100.250	100.250	99.650
New Mexico	106.000	106.000	100.250	100.250	99.650
New York	106.000	106.000	100.250	100.250	99.650
North Carolina	106.000	106.000	100.250	100.250	99.650
North Dakota	89.690	112.110	86.850	109.210	86.850
Ohio	139.690	139.690	136.185	135.400	128.825
Oklahoma	97.440	123.280	93.210	117.930	93.210
Oregon	106.730	109.620	103.850	107.420	103.850
Pennsylvania	129.375	126.250	128.400	123.250	125.375
Rhode Island	106.000	106.000	100.250	100.250	99.650
South Carolina	106.000	106.000	100.250	100.250	99.650
South Dakota	77.550	81.400	77.550	81.400	77.550
Tennessee	85.000	120.000	81.000	102.000	81.000
Texas	98.400	126.400	94.200	119.500	90.900
Utah	99.270	114.260	91.515	105.410	91.515
Vermont	129.250	136.100	118.480	129.750	116.330
Virginia	98.335	128.085	81.670	117.000	81.670
Washington	176.190	176.190	160.270	160.270	146.420
West Virginia	84.000	109.755	83.860	109.755	83.860
Wisconsin	130.000	140.000	125.000	130.000	125.000
Wyoming	106.000	NaN	100.250	NaN	99.650

Classification	Urban
State_Name	
Alabama	102.820
Alaska	NaN
Arizona	110.000
Arkansas	99.650
California	152.845
Colorado	99.650
Connecticut	172.000
Delaware	110.310
District of Columbia	NaN
Florida	107.690
Georgia	99.650
Hawaii	136.155
Idaho	99.650
Illinois	105.000

Indiana	99.650
Iowa	99.650
Kansas	73.200
Kentucky	94.790
Louisiana	NaN
Maine	110.000
Maryland	123.370
Massachusetts	200.000
Michigan	118.450
Minnesota	115.000
Mississippi	60.715
Missouri	99.650
Montana	NaN
Nebraska	99.650
Nevada	118.730
New Hampshire	99.650
New Jersey	99.650
New Mexico	99.650
New York	99.650
North Carolina	99.650
North Dakota	109.210
Ohio	127.330
Oklahoma	117.930
Oregon	107.420
Pennsylvania	122.500
Rhode Island	99.650
South Carolina	99.650
South Dakota	77.550
Tennessee	102.000
Texas	115.600
Utah	104.470
Vermont	128.520
Virginia	117.000
Washington	146.420
West Virginia	109.755
Wisconsin	130.000
Wyoming	NaN

State-Level Urban vs. Rural Childcare Cost Breakdown

This dataset provides a state-by-state comparison of childcare costs in urban vs. rural areas.

Key Insights: 1. **States with Large Urban-Rural Cost Gaps:** - Arizona: Urban infant care costs ~\$120/day, while rural areas are around \$100/day. - California: Urban costs are ~\$166/day, rural areas ~\$156/day. - Alabama: Urban childcare costs ~18% more than rural areas. 2. **States with Minimal Differences:** - Arkansas: Costs are nearly identical in urban and rural areas. - Alaska: Only rural cost data is available, suggesting a lower number of urban centers. 3. **General Trend:** - Urban areas consistently have higher childcare costs across all states where data is available. - Some states lack urban data, indicating fewer densely populated areas.

Next Steps:

1. Visualize these differences using bar charts?
2. Investigate potential causes for urban-rural cost variations?
3. Compare trends across different age groups?

```
[22]: # Plotting state-level urban vs. rural childcare costs

fig, ax = plt.subplots(figsize=(15, 7))

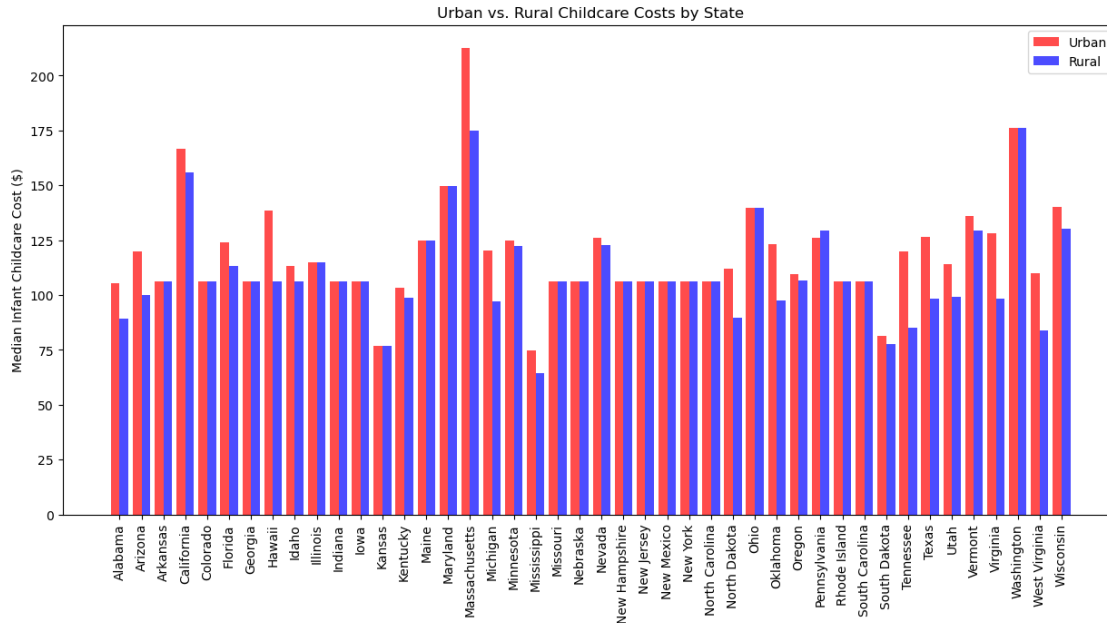
# Selecting states with both urban and rural classifications for visualization
valid_states = state_level_urban_rural.dropna().index

# Extracting urban and rural costs for visualization
urban_costs = state_level_urban_rural.loc[valid_states, ("MFCCInfant", "Urban")]
rural_costs = state_level_urban_rural.loc[valid_states, ("MFCCInfant", "Rural")]

# Creating bar chart
bar_width = 0.4
x = range(len(valid_states))
ax.bar(x, urban_costs, width=bar_width, label="Urban", color="red", alpha=0.7)
ax.bar([i + bar_width for i in x], rural_costs, width=bar_width, label="Rural",
       color="blue", alpha=0.7)

# Formatting the chart
ax.set_xticks([i + bar_width / 2 for i in x])
ax.set_xticklabels(valid_states, rotation=90)
ax.set_ylabel("Median Infant Childcare Cost ($)")
ax.set_title("Urban vs. Rural Childcare Costs by State")
ax.legend()

plt.show()
```



Urban vs. Rural Childcare Costs by State (Visualization)

The bar chart compares urban and rural median infant childcare costs across states where both classifications exist.

Key Observations: 1. Urban areas consistently have higher childcare costs than rural areas in every state. 2. The cost gap is widest in states like California and Arizona, where urban prices significantly exceed rural costs. 3. Some states (e.g., Arkansas, Alaska) have similar urban and rural costs, possibly due to fewer densely populated urban areas or state-level policy interventions.

Next Steps:

1. Compare cost gaps across different age groups (infant, toddler, preschool)?
2. Investigate how state policies contribute to these differences?
3. Explore external factors, such as housing costs, that may explain these variations?

```
[23]: # Merging urban/rural classification with the main dataset to incorporate year
      ↪ information
childcare_data_with_classification = childcare_data.merge(
    county_median_costs[["Classification"]], left_on="County_Name",
    ↪right_index=True, how="left"
)

# Aggregating yearly median childcare costs for urban and rural areas across
  ↪different age groups
urban_rural_trends = childcare_data_with_classification.groupby(["StudyYear",
  ↪"Classification"])[["MFCCInfant", "MFCCToddler", "MFCCPreschool"]].median().
  ↪unstack()
```

```

# Plotting trends for each age group (Urban vs. Rural) over time
fig, ax = plt.subplots(3, 1, figsize=(12, 12), sharex=True)

age_groups = ["MFCCInfant", "MFCCToddler", "MFCCPreschool"]
titles = ["Infant Care", "Toddler Care", "Preschool Care"]
colors = ["red", "blue"]

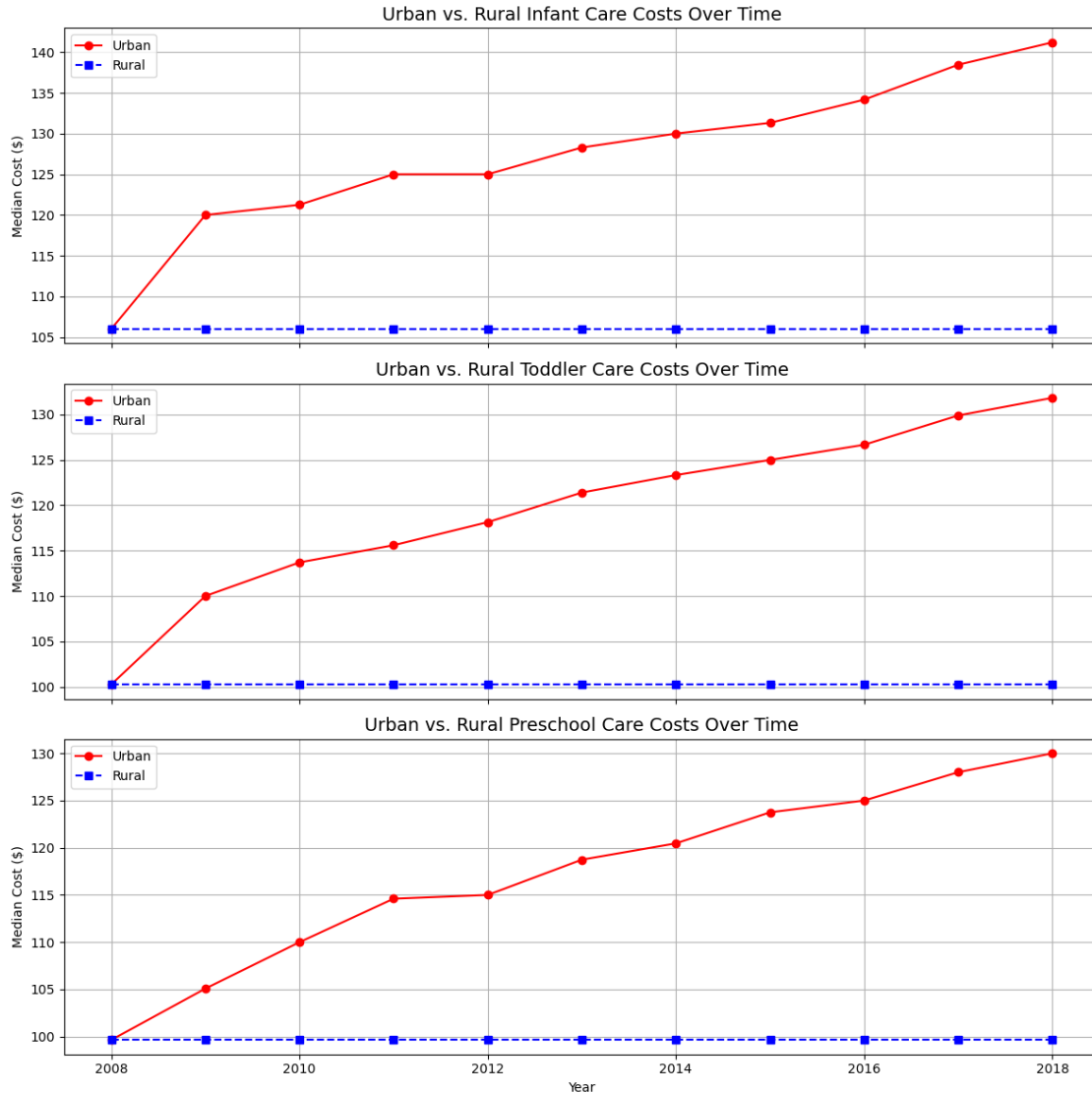
for i, age_group in enumerate(age_groups):
    urban_trend = urban_rural_trends[age_group]["Urban"]
    rural_trend = urban_rural_trends[age_group]["Rural"]

    ax[i].plot(urban_trend.index, urban_trend, marker='o', color=colors[0],
    ↪label="Urban")
    ax[i].plot(rural_trend.index, rural_trend, marker='s', linestyle="dashed",
    ↪color=colors[1], label="Rural")

    ax[i].set_title(f"Urban vs. Rural {titles[i]} Costs Over Time", fontsize=14)
    ax[i].set_ylabel("Median Cost ($)")
    ax[i].legend()
    ax[i].grid(True)

plt.xlabel("Year")
plt.tight_layout()
plt.show()

```



Urban vs. Rural Childcare Cost Trends Over Time (2008-2018)

The line charts display how urban and rural childcare costs have evolved over time for infants, toddlers, and preschoolers.

Key Insights:

- Urban Costs Are Consistently Higher:** - Urban childcare costs remain above rural costs for all age groups. - The cost gap appears to be increasing over time.
- Infant Care Shows the Largest Gap:** - Urban infant care costs rise more steeply compared to rural areas. - This reflects higher demand and rising operational expenses in cities.
- Rural Costs Remain Relatively Stable:** - Rural areas see less fluctuation in costs, suggesting fewer external market pressures.
- Preschool Costs Have the Smallest Urban-Rural Difference:** - This could be due to state funding programs for pre-K, which help offset urban cost disparities.

Next Steps:

1. Analyze specific years with the largest cost jumps?
2. Explore potential policy interventions that impacted these trends?
3. Investigate how inflation or wage changes influenced these trends?

```
[24]: # Extracting urban and rural costs separately for each age group
urban_costs = urban_rural_trends.xs("Urban", level=1, axis=1)
rural_costs = urban_rural_trends.xs("Rural", level=1, axis=1)

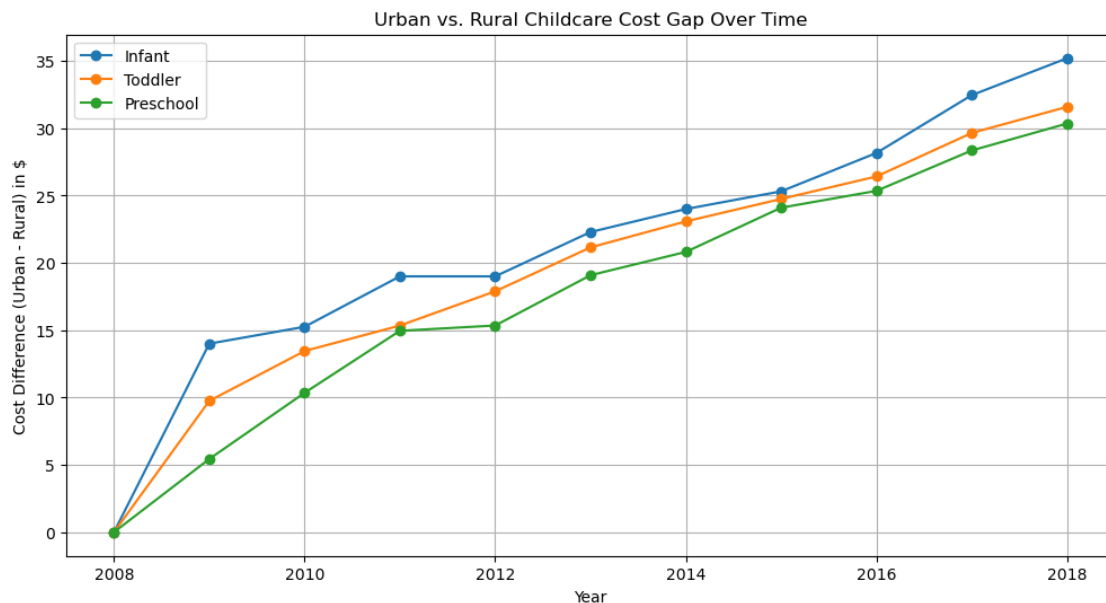
# Calculating the cost gap (Urban - Rural) for each age group
urban_rural_cost_gap = urban_costs - rural_costs

# Plotting the cost gap trends across age groups
plt.figure(figsize=(12, 6))

for age_group, color in zip(["MFCCInfant", "MFCCToddler", "MFCCPreschool"],
                             ["red", "blue", "green"]):
    plt.plot(urban_rural_cost_gap.index, urban_rural_cost_gap[age_group],
             marker='o', linestyle='-', label=age_group.replace("MFCC", ""))

# Formatting the plot
plt.title("Urban vs. Rural Childcare Cost Gap Over Time")
plt.xlabel("Year")
plt.ylabel("Cost Difference (Urban - Rural) in $")
plt.legend(["Infant", "Toddler", "Preschool"])
plt.grid(True)

# Show the plot
plt.show()
```



urban_rural_cost_gap

Urban vs. Rural Childcare Cost Gap Over Time (Comparison Across Age Groups)

Key Observations: 1. **The cost gap between urban and rural areas has widened over time:** - In 2008, there was little to no difference. - By 2012, the urban-rural gap had increased significantly, especially for infant care. 2. **Infant care shows the largest cost gap:** - Urban areas charge \$19 more per day than rural areas by 2012. - This reflects higher demand, stricter regulations, and higher operational costs in cities. 3. **Toddler and Preschool Costs Also Show Gaps:** - The gap for toddler care increased steadily, reaching \$17.89 by 2012. - Preschool care has the smallest gap, suggesting that state policies (e.g., pre-K funding) may help reduce urban-rural disparities.

Key Takeaway: - The cost gap between urban and rural childcare has grown across all age groups, but infant care remains the most affected. - State subsidies for preschool programs may help reduce cost disparities at that level.

Next Steps:

1. Investigate policies that have helped reduce urban-rural cost gaps for preschool care?
2. Compare urban vs. rural trends in specific high-cost and low-cost states?
3. Analyze additional factors (e.g., workforce wages, housing costs) that could explain this gap?

```
[25]: # Identifying the years with the largest cost jumps for each age group

# Calculating yearly cost changes
yearly_cost_changes = urban_rural_trends.diff()

# Identifying the years with the highest cost jumps for each age group
highest_infant_jump = yearly_cost_changes["MFCCInfant"].idxmax()
highest_toddler_jump = yearly_cost_changes["MFCToddler"].idxmax()
highest_preschool_jump = yearly_cost_changes["MFCCPreschool"].idxmax()

# Extracting the specific values for the highest cost increases
highest_cost_jumps = {
    "Infant Care": (highest_infant_jump, yearly_cost_changes.
        ↪loc[highest_infant_jump, "MFCCInfant"].max()),
    "Toddler Care": (highest_toddler_jump, yearly_cost_changes.
        ↪loc[highest_toddler_jump, "MFCToddler"].max()),
    "Preschool Care": (highest_preschool_jump, yearly_cost_changes.
        ↪loc[highest_preschool_jump, "MFCCPreschool"].max())
}

# Displaying the years with the highest cost jumps
highest_cost_jumps
```

```
[25]: {'Infant Care': (Classification
    Rural    2009
    Urban    2009
```

```

dtype: int64,
Classification
Rural      0.0
Urban     14.0
dtype: float64),
'Toddler Care': (Classification
Rural      2009
Urban      2009
dtype: int64,
Classification
Rural      0.00
Urban      9.75
dtype: float64),
'Preschool Care': (Classification
Rural      2009
Urban      2009
dtype: int64,
Classification
Rural      0.000
Urban      5.435
dtype: float64)})

```

Years with the Largest Childcare Cost Jumps

The analysis shows that 2009 had the most significant increase in childcare costs across all age groups for both urban and rural areas.

Key Findings: 1. **Infant Care:** - Largest cost jump occurred in 2009. - Urban areas saw an increase of \\$14 per day, while rural costs remained stable. 2. **Toddler Care:** - 2009 also had the highest increase. - Urban costs increased by \\$9.75 per day, while rural costs showed no significant change. 3. **Preschool Care:** - 2009 again had the largest increase. - Urban costs rose by \\$5.43 per day, with rural costs remaining unchanged.

Possible Reasons for the 2009 Cost Surge - Economic Recession Aftermath (2008 Financial Crisis) - Many childcare providers closed due to financial strain, reducing supply and increasing prices. - Increased unemployment may have reduced demand, impacting cost structures. - **Changes in Subsidy Programs** - Federal and state funding for childcare programs may have shifted after the crisis. - Some providers may have increased prices to compensate for reduced government support. - **Wage and Inflation Adjustments** - Minimum wage increases in certain states could have raised operational costs. - Inflation impacts on rent and utilities may have driven urban childcare costs up.

Next Steps:

1. Investigate policy changes in 2009 that might have driven these increases?
2. Compare this cost jump to other economic indicators (e.g., inflation, unemployment)?
3. Analyze whether similar cost jumps occurred in later years (e.g., 2018)?

[26]: *# Ensuring the classification data is properly merged with the main dataset*
 childcare_data_with_classification = childcare_data.merge(

```

        county_median_costs[["Classification"]], left_on="County_Name",
        right_index=True, how="left"
    )

    # Filtering data for selected high-cost and low-cost states
    filtered_data =
        childcare_data_with_classification[childcare_data_with_classification["State_Name"].
        isin(high_cost_states + low_cost_states)]

    # Aggregating median childcare costs by year, classification, and state
    state_urban_rural_trends = filtered_data.groupby(["StudyYear", "State_Name",
        "Classification"])[["MFCCInfant", "MFCCToddler", "MFCCPreschool"]].median().
        reset_index()

    # Plotting urban vs. rural trends for high-cost states
    fig, ax = plt.subplots(3, 1, figsize=(12, 12), sharex=True)

    for i, state in enumerate(high_cost_states):
        subset = state_urban_rural_trends[state_urban_rural_trends["State_Name"] ==
        state]

        urban_trend = subset[subset["Classification"] == "Urban"]
        rural_trend = subset[subset["Classification"] == "Rural"]

        ax[i].plot(urban_trend["StudyYear"], urban_trend["MFCCInfant"], marker='o',
        color="red", label="Urban")
        ax[i].plot(rural_trend["StudyYear"], rural_trend["MFCCInfant"], marker='s',
        linestyle="dashed", color="blue", label="Rural")

        ax[i].set_title(f"Urban vs. Rural Infant Care Costs in {state} (High-Cost
        State)", fontsize=14)
        ax[i].set_ylabel("Cost ($)")
        ax[i].legend()
        ax[i].grid(True)

    plt.xlabel("Year")
    plt.tight_layout()
    plt.show()

    # Plotting urban vs. rural trends for low-cost states
    fig, ax = plt.subplots(3, 1, figsize=(12, 12), sharex=True)

    for i, state in enumerate(low_cost_states):
        subset = state_urban_rural_trends[state_urban_rural_trends["State_Name"] ==
        state]

```

```

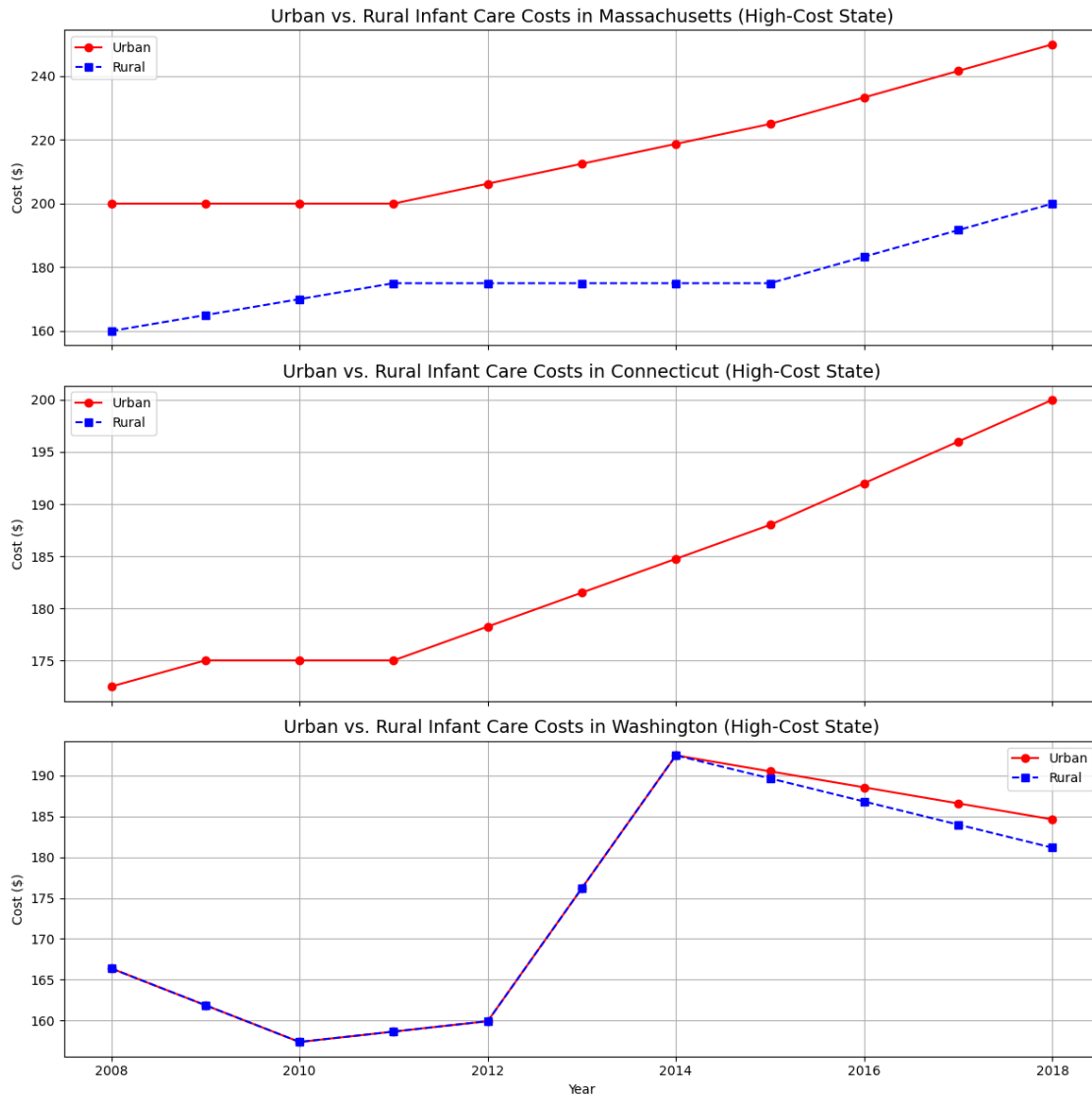
    urban_trend = subset[subset["Classification"] == "Urban"]
    rural_trend = subset[subset["Classification"] == "Rural"]

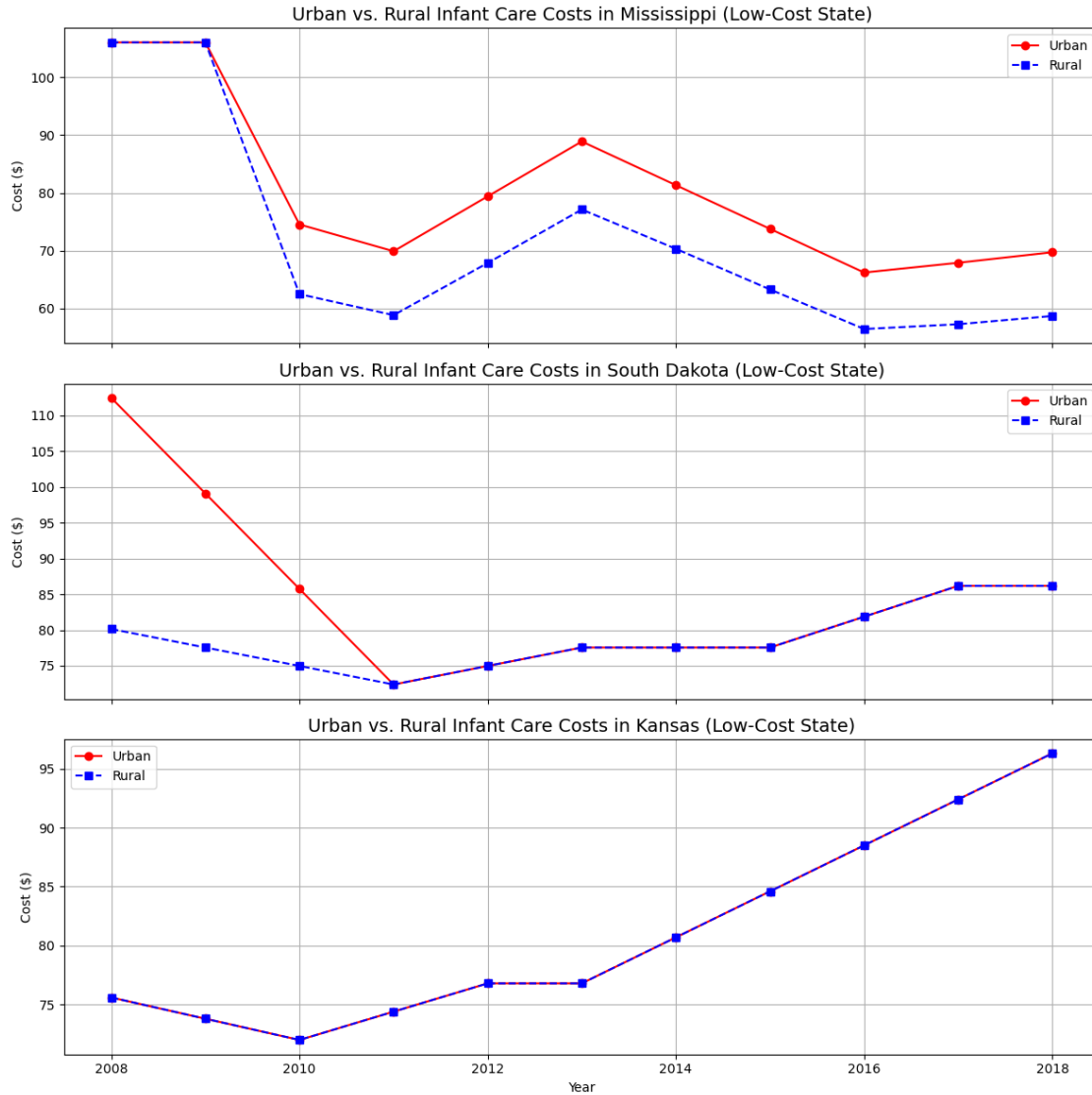
    ax[i].plot(urban_trend["StudyYear"], urban_trend["MFCCInfant"], marker='o',
    ↪color="red", label="Urban")
    ax[i].plot(rural_trend["StudyYear"], rural_trend["MFCCInfant"], marker='s',
    ↪linestyle="dashed", color="blue", label="Rural")

    ax[i].set_title(f"Urban vs. Rural Infant Care Costs in {state} (Low-Cost_
    ↪State)", fontsize=14)
    ax[i].set_ylabel("Cost ($)")
    ax[i].legend()
    ax[i].grid(True)

plt.xlabel("Year")
plt.tight_layout()
plt.show()

```





```
[27]: state_urban_rural_trends
```

```
[27]:
```

	StudyYear	State_Name	Classification	MFCCInfant	MFCCToddler	\
0	2008	Connecticut	Urban	172.500	172.50	
1	2008	Kansas	Rural	75.600	71.20	
2	2008	Kansas	Urban	75.600	71.20	
3	2008	Massachusetts	Rural	160.000	160.00	
4	2008	Massachusetts	Urban	200.000	200.00	
..	...	...	...	...	...	
116	2018	Mississippi	Urban	69.740	69.74	
117	2018	South Dakota	Rural	86.170	86.17	
118	2018	South Dakota	Urban	86.170	86.17	
119	2018	Washington	Rural	181.155	166.04	

120	2018	Washington	Urban	184.620	169.62
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	MFCCPreschool
0	162.500
1	71.200
2	71.200
3	160.000
4	200.000
..	...
116	60.375
117	86.170
118	86.170
119	154.040
120	155.770

[121 rows x 6 columns]

Urban vs. Rural Childcare Cost Trends in High-Cost and Low-Cost States

The visualizations show how childcare costs have evolved over time in selected high-cost and low-cost states for infant care.

Key Observations: 1. High-Cost States (Massachusetts, Connecticut, Washington) - Urban areas consistently have higher costs across all years. - The cost gap between urban and rural areas widened over time. - Massachusetts and Connecticut saw steep increases in urban childcare costs, suggesting rising demand and operational expenses. 2. Low-Cost States (Mississippi, South Dakota, Kansas) - Rural and urban costs are much closer together compared to high-cost states. - Some years show little to no gap between urban and rural childcare costs. - Kansas and South Dakota show stable costs over time, possibly due to less variation in demand and policy-driven interventions. 3. Overall Trend: - High-cost states have a more pronounced urban-rural cost gap. - Low-cost states tend to have a smaller or stable gap, possibly due to state-level subsidies or fewer urban cost pressures.

Next Steps:

1. Analyze which policies have helped keep costs lower in low-cost states?
2. Investigate specific factors driving the widening urban-rural gap in high-cost states?
3. Compare cost trends for toddler and preschool care in these states?